Inherent Safety Principles

- Minimization, Isolation, Downsizing, Substitution, Moderation, Simplification
- Engineering Controls

Hazards Analysis

HAZOP columns & guide words

FMEA columns

FTA

- Define Process
- Get Process Safety Info (PSI)
- Understand PSI
- Identify top level events & sub events & sub sub events
- Complete Table
- Issue Safety Action Requests

DHA vs PHA

process or design hazard analysis

FMEA / FTA or HAZOP

with summary & action items

Logic & Safety

Deductive vs. Inductive Reasoning

↳ General → Specific

Validity - Valid Arguments & Conclusions

Logical Fallacies

AI Tools - How does the algorithm of AI work?

forward vs. backward propagation

Industrial Hygiene

TWA
PEL

Toxicity

Responses to Toxicants

- Irreversible vs. Reversible

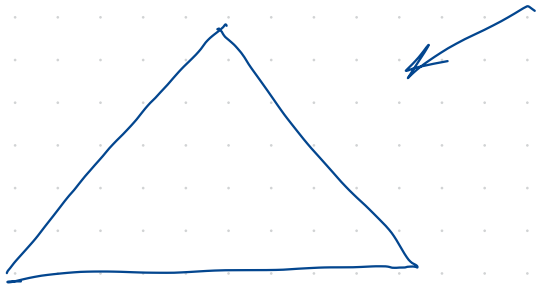
How are toxins eliminated from the body?

Chemical Reactivity & ΔG

Initiation Mechanisms

- ESD
- Thermal (Arrhenius)
- Friction
- Impact
- Flame

Flammability



VOL, LFL, UFL, LOL, LOC, etc.

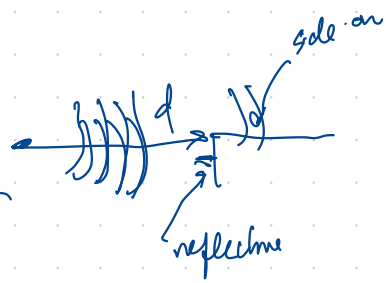
Air Line, Stoichiometric Line

Simple LFL, UFL estimates

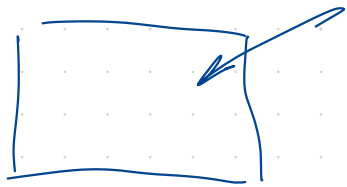
Explosions Mouse effect

TNT equivalence

Side-on overpressure



Balances



Concentration as a function of time for non-steady state scenarios

what if Temp isn't constant?

Orifice Flow

How fast does a gas or liquid exit an orifice or hole in the structure?

How Long 3-hours

1 sheet of paper 8x11

1 Calculator

Testing center 4-10 of March

Whoosh Bottle

$$\frac{dn}{dt} = \dot{n}_{in} - \dot{n}_{out}$$

Mach #

Eq. for gas flow

$$\dot{n}_{out} = P_0 A \, \Gamma_M$$

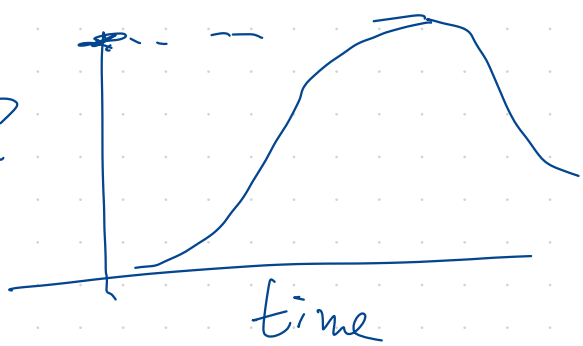
$$r = k P_E^{m} P_H^{n} \leftarrow \text{kinetics}$$

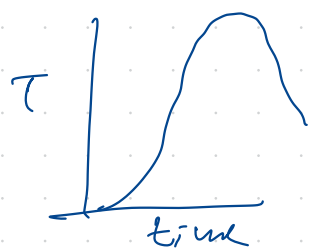
$$\frac{dn_{CO_2}}{dt} = \underline{\qquad\qquad}$$

$$H = U + P\bar{V} = \underline{U + RT}$$

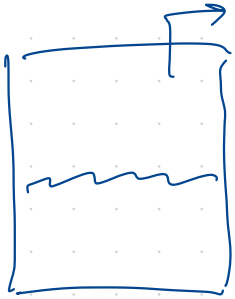
combustion properties

$$\frac{d(nU)}{dt} = n\frac{dU}{dt} + U\frac{dn}{dt} = \underline{\dot{n}_{in}\bar{H}_{in}} - \underline{\dot{n}_{out}\bar{H}}$$





$$PV = nRT$$



Gas scenario

$$V\frac{dP}{dt} + \cancel{P\frac{dV}{dt}}^{\,0} = Rn\cancel{\frac{dT}{dt}}^{\,0} + RT\frac{dn}{dt}$$

$$\frac{dn}{dt} = -\dot{n}_{out}$$

$$\frac{dP}{dt} = -\frac{RT}{V}\dot{n}_{out}$$

Isothermal

Adiabatic